

MethylPhys CPG - Operations Manual, chain v3

Development build, not commissioned; not a diagnostic test. Generated 2026-10-04 by manual/build_manual_v3.py from the files named at the start of each part. The procedure is sop/MethylPhys_CPG_SOP_v3.md.

Part 1. Operations - chain v3, neutrophils

Source: manual/OM_v3_neutrophil_chain.md

Operations Manual — chain v3, neutrophils (operator chapter)

Build: development v3 (2026-10-01), DEVELOPMENT - not commissioned; this chapter re-checked against the code on 2026-10-03 and updated for development round 2 on 2026-10-04; Stage T updated on 2026-10-04 for self-tare II then the median tare, adopted by the author (boxruns/run1/JOBS.md job A; doors/DEV_SELFTARE_02.md; development log 2026-10-04, DEV-PAIRED-01). Until then Stage T was the median tare alone (DEV-TARE-02, 2026-10-02). Self-tare II was wired into Stage T on 2026-10-04 (development log DEV-SELFTARE-03). The procedure is in sop/MethylPhys_CPG_SOP_v3.md; this chapter covers running it and reading the output. The PDF manual in this folder (MethylPhys_CPG_Operations_Manual.pdf) is built from this chapter, the generated chain sequence and the toolkit list by build_manual_v3.py. The class-floor engine (v2) and its manual were retired on 2026-10-03 and are archived privately. Paths are relative to Biological_Physics/MethylPhys/.

Before a run

1. Use EPIC v1 IDAT pairs (Grn and Red) of whole blood or isolated / sorted / purified neutrophils. Any other specimen (PBMC, other sorted fractions, bone marrow, cell lines, tissue, unspecified) is refused at intake with a report. Sex and age are optional: give them when known and they are recorded; with a declared sex Stage 0.8 compares it with the array.
2. Put at least 3 healthy reference specimens of the same specimen type on the same slide (else the same batch), run the same way, so each reading can be tared. This applies to whole blood and to isolated neutrophils. Stage T is self-tare II (each array tared on its own fixed sites; no references needed), then the median tare against these references.
3. Use Python 3.11 with the versions in chain/requirements.txt (methylprep 1.7.1, numpy 1.26.4, pandas 1.5.3, scipy 1.17.1; reportlab for the manual): `pip install -r chain/requirements.txt`.
4. Stage 1 needs the Illumina manifest. methylprep downloads it on first use into `$HOME/.methylprep_manifest_files/` (HOME must be writable; network access to `array-manifest-files.s3.amazonaws.com`); offline, place the files there by hand (doors/RUNBOOK.md section 1). In a batch, run the first array alone so the download is not raced.
5. The frozen inputs are in the repository under chain/Runtime Matrices/ (Met_A_Floors/, IAM_A_Positions/, Intake/); their hashes are in chain/FROZEN_INPUTS_v3.json.

Run

From chain/MethylPhys_Interface/ (run on the box: Stage 1 needs the manifest):

```
python run_sample.py --grn <Grn> --red <Red> --engine v3 --specimen "whole blood" --array-type EPIC_v1 \
--id <id> --out <id>.html # add --sex <F|M> --age <years> when known
```

- Isolated neutrophils: `--specimen "isolated neutrophils"`.
- Pass 2 (the tare): run each specimen again, whole blood and isolated alike, with `--slide-ref-A a1,a2,a3`, where those are the references' A values from pass 1 (met_a.A, self-tared, before the median tare), or with `--slide-ref-table refs.csv` (column A, optional column id; a row whose id is the specimen's own --id is left out). Fewer than 3 references: the median tare does not run and the reading stays untared.

- Self-tare II (Stage T step 1, adopted 2026-10-04): always runs (wired into Stage T on 2026-10-04); no flag is needed. It maps each probe design onto the reference arrays' scale by this array's own low and high fixed-site anchors before Met-A is read (tare.selftare_ii). The printed A_rel is the median tare of the self-tared A. The noise index reads the betas before self-tare II. --dev-selftare-ii is a no-op alias, kept so recorded commands still run.
- A beta table already calibrated by this chain's Stage 1 (two-column CSV cpq_id, beta; Stage 0 does not run):
python run_sample.py --betas <table>.csv --specimen "whole blood" --id <id> --out <id>.html.
- Second draw of the same person (stage 12b): run the first draw with --save-betas <id>_betas.parquet --patient-id <hash>; run the second with the same --patient-id plus --prior-betas <id>_betas.parquet --prior-bundle <id>_bundle.json.
- Identifiers: the bundle and the ledger carry the sha256 hash of --id; the report keeps the id you typed.
- Development flags (DEVELOPMENT - not commissioned; never part of the reading; --dev-selftare-ii is a no-op alias since the adopted Stage T step 1 was wired in on 2026-10-04): --dev-selftare-ii --dev-direction --dev-trace --dev-foreign --dev-brightness (no extra input); --dev-nilc --dev-atlas-e --dev-percell-b --dev-sky with --atlas-v2 <IAMAtlas_v2.parquet> (--dev-sky needs healpy); --dev-epic-v2 --sesame-rscript <Rscript> for an EPIC v2 IDAT pair. Each writes development.<stage> into the bundle and a report section.
- Sequencing (Stage Q, IAM-A): python run_sample.py --pat <file>.pat.gz --id <id> --out <id>.html, or python run_sample.py --site-table <sites>.csv --seq-pipeline loyfer_pat_v1 --id <id> --out <id>.html.

Output: <id>.html, <id>_bundle.json beside it, and one row appended to evidence_ledger.jsonl in the same folder (--ledger to change; --no-bundle writes neither the bundle nor the row). Exit 0 when a report is written (a refusal still writes one); exit 2 on a Stage 0 QUARANTINE, with no report, no bundle and no ledger row (a command-line error also exits 2).

Batch runners (written for the compute box; they read box paths and roster files and are not general tools):
chain_tests/run_chain_acceptance.py (pass 1 every specimen; pass 2 re-runs each whole-blood specimen with the other specimens of its group as references; isolated specimens are not re-run) and
doors/data/DEV_REPL_V3_01_run/run_proc_repl_v3_01.py (pass 2 with --slide-ref-table built from the other arrays on the same slide, else the batch; exact commands in COMMANDS.md beside it).

Checks: python chain/release_check_v3.py (or python kit/release_check.py) - exit 0 only when every check passes (F1, F1b, S1-S4, E1-E10, M1). The IDAT checks E1-E3 and E6 need the manifest (run on the box). Manual: python manual/build_manual_v3.py rebuilds the PDF.

Reading the report

field	meaning
Refused (specimen)	SPECIMEN_REFUSED: the specimen has no reference in chain v3; the refusal names it. No reading, Stage 1 does not run
Refused	the platform check: not an EPIC v1 vector (array type, EPIC v2 probe names, or 700,000 probes or fewer); an EPIC v2 IDAT pair is refused at intake, before Stage 1. No reading
Stage 0 intake	verdict PROCEED / PROCEED_WITH_PENALTY / QUARANTINE (QUARANTINE produces no report), call rate, flags; not run for --betas, --pat, --site-table or --no-intake
Stage 1	poobah detection, call rate and controls: recorded, not gated
Stage A composition	the 8 blood groups (groups at 1 % or more are listed). Whole blood is read when neutrophils are ≥ 20 % and ≥ 867 of the 963 markers are measured
Stage M Met-A	isolated cells: A against the own floor, drawn on the gauge as untared until tared. Whole blood untared: A is a number only, no gauge position. Shift per 1 % loss of the neutrophil pattern

field	meaning
Stage T tare	adopted 2026-10-04: self-tare II ($\beta' = L_r + (\beta - L)(U_r - L_r)/(U - L)$) per probe design, from this array's own fixed-site anchors L, U and the reference arrays' L _r , U _r ; nothing fitted), then the median tare. Printed: A _{rel} = A ÷ median of the references (self-tared A; self-tare II wired in on 2026-10-04, record tare . selftare . ii); number of references and their median; reference spread; detection limit (% loss of the pattern) = 2 × spread ÷ shift per 1 % loss. Normal = 0.95–1.05
Noise index N	mean H(β) on the 48,528 noise sites. Above 0.149 on an untared reading: state withheld, A printed as a number. Fewer than 90 % of the noise sites measured: N is not formed and the state is withheld, tared or not, with the counts and the reason
Methylated sites mean β	below 0.5: past the entropy ceiling; read β , not A
Stage MC C-score	genomic clustering of the departures (healthy = 1), with the healthy held-out range. Development: no band yet
Stage Q IAM-A	sequencing only: IAM-A, copy error eps, position P, eps0, the two halves, opportunities; the IAM-A C-score (development: independent copy errors give 1; band not set)
Development stages	only with a development flag: one row per flagged stage, labelled DEVELOPMENT - not commissioned
Withheld	what the build does not print, and why
Stage 12b difference map	only with --prior-betas/--prior-bundle: per-address difference to an earlier draw of the same person, or the refusal naming what differs (identifier hash, array type, pipeline)
Red flags	STOP / WITHHELD / CAUTION / NOTE, from the bundle; also written to the bundle as red_f_lags
Safeguards	rendered-claim scan (no disease, cohort or population words in the prose), formula self-test, anchors (Normal band, floor, C baseline, N _{max} read from the frozen files), deconvolver conformance, atlas separability (NOT_RUN until stage 3 is wired)
Troubleshooting, Integrity, Chain file inventory, Run it yourself	what to do for each condition on this specimen; IDAT hashes and chain commit; every frozen input and module with its hash; the exact command
The chain, Toolkit stages	SOP 2b stage table; each toolkit stage PASS / FAIL / REFUSED / NOT_RUN / NOT_BUILT on this specimen

Faults

symptom	cause	action
QUARANTINE_INCOMPLETE_MANIFEST	array type not declared and not readable from the IDAT header (sex and age are optional since 2026-10-04)	supply --array-type
SPECIMEN_REFUSED	the specimen is not whole blood or isolated / sorted / purified neutrophils	none: that specimen needs its own reference first
withheld: only <n> of the 48,528 noise sites were measured	fewer than 90 % of the noise sites passed Stage 1 detection	re-hybridise or check the array's signal; the state cannot be shown without the array's own noise
ENVIRONMENT_MISSING_MANIFEST (exit 3; before 2026-10-03 this showed as QUARANTINE_CORRUPT_IDAT) with PermissionError or a connection error on .methylnprep_manifest_files	the manifest is not in its cache and could not be downloaded; the IDAT is not at fault and is not judged	make HOME writable and allow the download, or place the manifest files by hand; re-run
Stage 0 QUARANTINE, other hard failures (call_rate, detection, ctrl_qc, sex, integrity, hm450_coverage, intake_deferred:...)	the array failed intake	none: no reading by rule; the flags name the check
neutrophil fraction <f> < 0.2: fraction reported, A withheld	neutrophils < 20 % of the whole blood	none: the fraction is reported and A is withheld by rule
only <n> of 963 composition markers measured / only <n> of 6000 ... sites measured	fewer than 90 % measured	none: A withheld

symptom	cause	action
untared: <n> same-run reference arrays (>= 3 required)	fewer than 3 references supplied	run the references, then pass them with --slide-ref-A or --slide-ref-table
tare.selftare_ii NOT_RUN, dev_selftare_typeII_EPIC_v1.json not found (the reading then uses β without self-tare II)	the self-tare II file is missing from chain/Runtime Matrices/Development/	restore it from the repository
withheld: noise index <N> > 0.149 and no same-run tare	the array is noisier than the reference arrays; the gauge is not drawn	tare it against same-run references
refusal ... (450K or incomplete vector) ... (450K neutrophil floor pending)	fewer than 700,001 probes after Stage 1 (a 450K table, or an EPIC array that lost probes at detection)	none: v3 reads EPIC v1 only. A 450K IDAT pair never gets here: it quarantines at Stage 0 (0.1 array-type mismatch when --array-type EPIC_v1 is given, else 0.7b coverage)
refusal ... (no frozen neutrophil floor for this platform)	declared array type is not EPIC_v1, or EPIC v2 probe names	none: no floor for that platform
IAM-A refusal position for neutrophils was measured on loyfer_pat_v1, not <pipeline>/too few opportunities	another read-level pipeline, or fewer than 100,000 opportunities	none: P is valid only for its own pipeline

Part 2. The chain, step by step

Source: doors/CHAIN_SEQUENCE.md

The chain, step by step - derived from the code

Generated by chain/build_chain_sequence.py from the AST of run_sample.py, conductor_v3.py and stage_q_iam_a.py. Do not edit by hand: re-run the generator. Every step below is a call the code actually makes, in the order it makes it. Chain v3 is the only engine; the class-floor engine (v2) was retired on 2026-10-03 and is archived privately.

The live path (chain v3) - 27 steps, in order

#	step	implemented in	what it does
1	specimen_refusal	stage_0_intake.py	None when intake accepts the specimen (ACCEPTED_SPECIMENS); otherwise the plain refusal text naming the specimen (decision L)
2	step_0_1_idat_arrival	stage_0_intake.py	SOP section 0.1_idat_arrival
3	step_0_2_manifest_creation	stage_0_intake.py	SOP section 0.2_manifest_creation
4	step_0_3_integrity_hash	stage_0_intake.py	SOP section 0.3_integrity_hash
5	step_0_4_control_probe_validation	stage_0_intake.py	SOP section 0.4_control_probe_validation
6	step_0_5_detection_pvalue_qc	stage_0_intake.py	SOP section 0.5_detection_pvalue_qc
7	step_0_6_bead_count_qc	stage_0_intake.py	SOP section 0.6_bead_count_qc
8	step_0_7_call_rate	stage_0_intake.py	SOP section 0.7_call_rate
9	step_0_7b_platform_coverage	stage_0_intake.py	SOP section 0.7b_platform_coverage
10	step_0_8_sex_check	stage_0_intake.py	SOP section 0.8_sex_check
11	step_0_9_decision_gate	stage_0_intake.py	SOP section 0.9_decision_gate
12	Stage 1 - IDAT calibration	stage_1_idat_calibration.py	Stage 1 - IDAT calibration to beta (SOP Stage 1, steps 1.1-1.2 + 1.5) _called by run_sample.py when given --grn/--red; skipped when given --betas_
13	platform_refusal	conductor_v3.py	None when the specimen is EPIC v1; otherwise the refusal text
14	stage_t_selftare_ii	conductor_v3.py	Stage T step 1, self-tare II (adopted 2026-10-04, DEV-SELFTARE-02): per probe design, $\beta' = Lr + (\beta - L)(Ur - Lr)/(U - L)$ from this
15	stage_m_isolated	conductor_v3.py	Isolated / sorted neutrophils against their own floor. The own-floor state is kept as state_own_floor; the reading's state is
16	stage_a_composition	conductor_v3.py	EPIC blood composition (blood_composition_EPIC_v1): 8 groups from Salas purified EPIC cells; markers exclude the neutrophil sites;
17	stage_m_blood	conductor_v3.py	Whole blood: $\text{Met-A} = \text{mean } H(\beta) / \text{mean } H(e)$ at the neutrophil sites, $e = \text{sum_g_f_g_mu_g}$ (EPIC purified group profiles)

#	step	implemented in	what it does
18	noise_index	conductor_v3.py	N = mean H(beta) over the noise sites measured on this array; None when fewer than MIN_NOISE_FRACTION are measured _stage 9: the array's noise index N over the noise sites_
19	stage_t_tare	conductor_v3.py	Same-run tare against >= MIN_REFS healthy references of the same specimen type run the same way (same slide, else same batch)
20	noise_gate	conductor_v3.py	Runtime Matrices/Met_A_Floors/noise_gate_EPIC_v1.json: N_max = top of the reference arrays' noise range (DEV-NOISE-01) _stage 9: N_max read here; run_neutrophil withholds the gauge state when N > N_max and the reading is untared_
21	stage_mc_cscore	conductor_v3.py	Met-A C-score (stage 6): clustering of the residual z map in genomic order - variance of the means of blocks of clustering_block
22	stage_q_iam_a.pat_site_table	stage_q_iam_a.py	Per-site table (pos, opp_A, err_A, opp_B, err_B) from one .pat(.gz) file, pipeline loyfer_pat_v1. pos = 'chrom:CpG index' _runs when run_sample.py is given --pat or --site-table_
23	stage_q_iam_a.read	stage_q_iam_a.py	site_table: per-site table (see module doc). cell: e.g. 'neutrophils'. pipeline: the read-level pipeline that produced the table _runs when run_sample.py is given --pat or --site-table_
24	stage_q_iam_a.cscore	stage_q_iam_a.py	IAM-A C-score for the pooled table (opp = opp_A + opp_B) and for each half. Returns {C, n_blocks, se_null, halves: {...}} _runs when run_sample.py is given --pat or --site-table_
25	serial_mode.check_same_person	serial_mode.py	Refuse unless patient hash, array type and pipeline match. Returns (ok, reason) _stage 12b: runs when run_sample.py is given --prior-betas and --prior-bundle_
26	serial_mode.delta_sky	serial_mode.py	beta_now - beta_prior per address on the addresses both draws measured. No expectation, no sigma, no composition _stage 12b: runs when run_sample.py is given --prior-betas and --prior-bundle_
27	Report	MethylPhys_Interface/repor t_v3.py	one self-contained HTML page plus the JSON bundle

Stage modules in chain/ that the live path does not call

module	what it implements	status
stage_2c_trace_detection.py	Stage 2c - is there any evidence of a trace class in this specimen?	module present; neither run_sample.py nor conductor_v3.py calls it (toolkit, chain/TOOLKIT.md)
stage_4_6_patient_cmb.py	Stage 4.6 — the patient's sky (PROC-CMB-01, 2026-09-21). Replaces the retired patient_brightness_comparison.py	module present; neither run_sample.py nor conductor_v3.py calls it (toolkit, chain/TOOLKIT.md)

module	what it implements	status
stage_a_composition_v2.py	Stage A - composition, atlas v2 solver (deconv_v2). Not called by chain v3 (conductor_v3.stage_a_composition uses blood_composition_EPIC_v1.json). Frozen settings (commissioned as PROC-DECONV-V2-01; specimen cell sets 2026-09-30)	module present; neither run_sample.py nor conductor_v3.py calls it (toolkit, chain/TOOLKIT.md)

Behind development flags (DEVELOPMENT - not commissioned; never part of a reading)

flag	step	implemented in	what it does
--dev-direction	dev_stages.direction_record	dev_stages.py	Signed move toward or away from 0.5 at the neutrophil identity sites against the cell's own reference, tared on same-run references (DEV-DIRECTION-02)
--dev-trace	dev_stages.trace_cell	dev_stages.py	3b: another blood group in an isolated-neutrophil specimen, called above 3 x this array's own standard error (DEV-TOOLKIT-ADDED-02)
--dev-foreign	dev_stages.foreign_cell	dev_stages.py	3c: a non-blood template in whole blood by NNLS with the blood groups, called above 3 x this array's own standard error (DEV-TOOLKIT-ADDED-02)
--dev-brightness	dev_stages.brightness	dev_stages.py	11b: 95 % interval on Met-A from the v3 per-site uncertainty (this array's own fixed-site noise), by Monte Carlo (DEV-TOOLKIT-ADDED-02)
--dev-nilc	dev_stages.nilc_e	dev_stages.py	Stage 4 NILC-e composition on the atlas_e templates (DEV-NILC-01 method), 8-group sums and the noise per fraction
--dev-atlas-e	dev_stages.atlas_e	dev_stages.py	Stage 3 atlas_e composition (12 array-measured circulating atlas v2 cells, DEV-ATLAS-EPIC-02 method), 8-group sums
--dev-percell-b	dev_stages.percell_b	dev_stages.py	Stage 5 B-cell Met-A against the composition-matched expectation on the development B-cell floor (DEV-PERCELL-01 method)
--dev-sky	dev_stages.sky	dev_stages.py	Residual sky with the within-chromosome block-shuffle null and the look-elsewhere statistic by simulation (DEV-SKY-02)
--dev-epic-v2	dev_stages.epicv2_calibrate	dev_stages.py	EPIC v2 IDAT pair -> beta Series on EPIC v1 probe names (replicate v2 probes of one CpG averaged). SeSAmE openSesame(prepare), pOOBAH-masked

The toolkit (chain/TOOLKIT.md) - built; each stage wired only after its check passed

stage (SOP 2b)	name	module	status
3	Atlas deconvolution	chain/deconv_v2.py	toolkit: not yet wired into chain v3
3b	Trace-cell detection	chain/dev_stages.py	toolkit: not yet wired into chain v3

stage (SOP 2b)	name	module	status
3c	Foreign-cell detection	chain/dev_stages.py	toolkit: not yet wired into chain v3
4	NILC component separation	chain/dev_stages.py	toolkit: not yet wired into chain v3
4	NILC helper: lineage splitter	chain/Lineage_Splitter/lin eage_splitter.py	toolkit: not yet wired into chain v3
10	Directional decomposition	chain/dev_stages.py	toolkit: not yet wired into chain v3
11	Sky map	chain/dev_stages.py	toolkit: not yet wired into chain v3
11b	Surface brightness	chain/dev_stages.py	toolkit: not yet wired into chain v3
12	Sky statistics	chain/sky_statistics.py	toolkit: not yet wired into chain v3
12b	Difference map	chain/serial_mode.py	wired into chain v3 (optional input; see chain/TOOLKIT.md)
5	Met-A per cell (B cells)	chain/dev_stages.py	toolkit: not yet wired into chain v3
8	Self-tare on type II fixed sites	chain/dev_stages.py	wired into chain v3 (optional input; see chain/TOOLKIT.md)
1	EPIC v2 calibration	chain/dev_stages.py	toolkit: not yet wired into chain v3

Part 3. The toolkit

Source: chain/TOOLKIT.md

chain/TOOLKIT.md - the toolkit: built, kept, entering chain v3 one stage at a time (2026-10-03: 12b wired; 2026-10-04, development round 2: stages likely to work behind development flags)

Every module below is in the repository and importable. **None of it is part of a reading**; stage 12b (difference map) is wired as an optional second-draw comparison and changes no reading. Since 2026-10-04 (author decision N, doors/DEV_FLAGS_01.md) the stages likely to work run behind development flags of `run_sample.py` (chain/dev_stages.py): each writes `bundle["development"] [<stage>]`, labelled DEVELOPMENT - not commissioned, and none changes the reading. Development round 2 records: doors/DEV_ROUND2_REPORT.md. Each enters chain v3 at commissioning, one stage at a time, after its own pre-registered check has passed on v3 and the outcome is recorded in doors/ (order and checks: ../sop/MethylPhys_CPG_SOP_v3.md section 2b). Every module carries the line *Toolkit: not yet wired into chain v3; enters the chain at commissioning with its own pre-registered check*. Stage numbers and names are those of SOP v3 section 2b. Paths are relative to Biological_Physics/MethylPhys/. The class-era engine these modules were first used with (chain v2) was retired on 2026-10-03 and is archived privately; nothing here imports it.

stage	name (SOP 2b)	module / file	what it does (method)	inputs	status						
3	Atlas deconvolution	chain/deconv_v2.py (Deconv V2); wrapper chain/stage_a_composition_v2.py; atlas_e (12 array-measured circulating cells) chain/dev_stages.py (atlas_e) with chain/dev_composition.py; development copies atlas/v2/postbuild/scripts/deconv_v2.py, methylphys_v2.py	one joint non-negative solve over the atlas v2 cell types, $f \geq 0$, $\sum f = 1$, inverse-variance weights iterated, markers chosen per cell against its nearest cells	the atlas v2 reference (parquet, built by atlas/v2/scripts; not stored in the repository), a beta vector on the Stage 1 scale	development flag --dev-atlas-e (2026-10-04). Checks: DEV-ATLAS-EPI-C-02 held-out truth outside the bars; DEV-COMPOSITION-TRUTH-02 (GSE77797, another laboratory, 450K): every group within 0.03 except granulocytes (0.041)						

stage	name (SOP 2b)	module / file	what it does (method)	inputs	status						
3b	Trace-cell detection	chain/ dev_stages.py (trace_cell, round 2: another blood group in an isolated-neutrophil specimen, line = 3 x this array's own standard error); class-era chain/stage_2c_trace_detection.py with chain/Runtimee Matrices/trace_detection_panel_v1.json kept out (line from other arrays)	weighted regression of the residual (beta - neutrophil profile) on each group's contrast at the 963 markers; weights from the v3 per-site uncertainty	a beta vector; Runtime Matrices/Development/dev_selftare_typeII_EPIC_v1.json (per-site uncertainty)	development flag --dev-trace (2026-10-04). DEV-TOOLKIT-ADDED-02: 82.5 % called at 5 % (bar 95 %); 97.9 % with pure spike material; 0/360 unspiked						

stage	name (SOP 2b)	module / file	what it does (method)	inputs	status						
3c	Foreign-cell detection	chain/dev_stages.py (foreign_cell, round 2: NNLS on the 8 groups + a non-blood template, line = 3 x this array's own standard error); class-era chain/toolkit_foreign_detection.py with detection_panel_v3.js on kept out (line from other arrays, class-era scale)	non-negative fit at the markers plus sites where the template differs from every blood group by ≥ 0.25	a beta vector; Runtime Matrices/Development/dev_foreign_placenta_EPIC_v1.json	development flag --dev-foreign (2026-10-04). DEV-TOOLKIT-ADDED-02: 100 % called at 5 %, but also 100 % with nothing added (zero offset 0.023-0.048); needs a same-run zero						
4	NILC component separation	NILC-chain/dev_stages.py (nilc_e) with chain/dev_comp_methods.py; class-era chain/nilc_celltype_deconvolve.py (NILCCelltypeDeconvolver) kept out	internal linear combination per cell type on the atlas_templates, constrained, noise per fraction printed	the atlas v2 parquet (not in the repository), a beta vector	development flag --dev-nilc (2026-10-04). DEV-NILC-01 truth bars outside; DEV-COMPOSITION-TRUTH-02: granulocytes 0.061, CD4T 0.036 outside. The class-era module is kept out (every truth bar outside in DEV-NILC-01)						

stage	name (SOP 2b)	module / file	what it does (method)	inputs	status						
4	NILC helper: lineage splitter	chain/Lineage_Splitter/lineage_splitter.py (LineageSplitter)	splits one compartment's mass among its member classes by a small NNLS on the contrast CpGs (abs(mean_a - mean_b) >= min_delta, weighted by the contrast), with a condition-number check; reports the compartment only when the split is ill-conditioned	atlas CSV, a beta vector, the compartment fraction and its members	helper of stage 4; not wired (stage 4 check failed)						
10	Directional decomposition	chain/dev_stages.py (direction_record, round 2, physics only); class-era chain/Runtime Matrices/Directional_Panel/bidirectional_decomposition.py with directional_panels_v1_0.json kept out (z against other arrays, a disease sign)	D = mean over identity sites of \	r - 0.5\	- \	beta - 0.5\	against the cell's own reference; > 0 toward disorder; tared on same-run references (direction when \	D_rell	> 2 x reference spread)	a beta vector, the frozen profiles	development flag --dev-direction (2026-10-04). DEV-DIRECTION-02: decitabine and NTX-301 12/12 toward disorder; replicates 56/63 no direction (bar 95%)

stage	name (SOP 2b)	module / file	what it does (method)	inputs	status						
11	Sky map	chain/dev_stages.py (residual_z, sky); class-era chain/stage_4_6_patient_cmb.py (patient_sky_sigma, render_plate); mapping chain/Runtime Matrices/Patient_CMB/build_healpix_mapping.py	residual $z = (\text{beta} - \text{sum } f_{\mu}) / \text{sigma}$ per CpG on HEALPix NSIDE 128 (RING) in genomic order	a beta vector, NNLS8 fractions, atlas v2 parent means and SDs, the mapping	development flag --dev-sky (2026-10-04, needs healpy). DEV-SKY-02: against the within-chromosome block-shuffle null, bands 2, 5, 6 inside 0.9-1.1; I 2-8 1.84, I 25-64 1.14, I 65-128 1.15						
11b	Surface brightness	chain/dev_stages.py (brightness, round 2: v3 per-site uncertainty); class-era chain/toolkit_surface_brightness.py (attach_brightness_ci) kept out (class archives)	95 % interval on Met-A by Monte Carlo of $\text{beta}_i + N(0, s_i^2)$, s_i from this array's own fixed sites	a beta vector, Runtime Matrices/Development/dev_selft_are_typeII_EPIC_v1.json	development flag --dev-brightness (2026-10-04). DEV-TOOLKIT-ADDED-02: half-width 0.003 against same-per son differences 0.032; 3.8 % of pairs inside (bar 95 %)						

stage	name (SOP 2b)	module / file	what it does (method)	inputs	status						
12	Sky statistics	chain/sky_statistics.py (masked_spectrum, bandpowers, shuffled_null); block-shuffle null and look-elsewhere by simulation chain/dev_stages.py (block_shuffle_order, sky_from_z); null suite chain/CPG_Null_Runner/cpg_null_runner.py; per-run tool registry chain/cmb_tools.py	masked pseudo angular power spectrum (healpy anafast), six bands; within-chromosome block-shuffle null (runs of 50 sites); look-elsewhere: max band ratio against 100 shuffles of the same array	a HEALPix residual map (stage 11)	development flag --dev-sky (2026-10-04). DEV-SKY-02: look-elsewhere rate 91% on 163 healthy arrays (bar 8.4%)						
12b	Difference map	chain/serial_mode.py (check_sam_e_person, delta_sky, delta_cells, trajectory)	per-address beta difference of two draws of one person on the addresses both measured, refused unless both draws share the identifier hash, array type and pipeline; per-cell A differences across draws (design doors/PROCESSED_01_PREG.md)	two runs (bundles and beta vectors) of one person	wired 2026-10-03 behind -prior-betas/--prior-bundle (DEV-TOOLKIT-ADDED-01 passed; release check E5): per-address difference + same-person check. Not built: the difference drawn as a sky; delta_cells/trajectory (class-era fields) not called						

stage	name (SOP 2b)	module / file	what it does (method)	inputs	status						
5	Met-A per cell (B cells)	chain/dev_stages.py (percell_b); floor chain/Runtime Matrices/Met_A_Floors/met_A_floors_v1_2_ALLCELLS_development.json	B-cell Met-A against the composition-matched expectation (atlas v2 parents x stage 2 fractions)	a beta vector, the atlas v2 parquet	development flag --dev-percell-b (2026-10-04). DEV-PE RCELL-01 and DEV-NEW CELL-01: the new-cell rule is not met (purified 16/22, replicate 0.037)						
8	Self-tare on type II fixed sites	chain/dev_stages.py (anchors, selftare_map, selftare_ii); chain/Runtime Matrices/Development/dev_selftare_typeII_EPIC_v1.json	affine map of each probe design onto the reference arrays' scale from this array's own low and high fixed-site anchors; Met-A re-read	a beta vector	wired 2026-10-04 as Stage T step 1 (chain/conductor_v3.py: stage_t_selftare_ii, before A and M; DEV-SEL FTARE-03); --dev-selftare-ii is a no-op alias since. DEV-SEL FTARE-02: self-tare II then median tare meets every bar (replicate SD 0.016, 62/63; other laboratories 49/49; floor 6/6) - adopted by the author as the tare on 2026-10-04						

stage	name (SOP 2b)	module / file	what it does (method)	inputs	status						
1	EPIC v2 calibration	chain/dev_stages.py (epic v2_calibrate); chain/dev_epicv2_sesame.R (SeSAR (SeSAR Me))	SeSAR on an EPIC v2 IDAT pair, v2 probes averaged into EPIC v1 names, read on the v1 floor at the shared sites (cross-version, development)	an EPIC v2 IDAT pair, an Rscript with sesame	development flag --dev-epic-v2 (2026-10-04). DEV-EPI C-V2-01: 72/72 calibrated; identity-site coverage 5,391 of 6,000; same-blood v2 - v1 SD 0.054; no v2 floor						

Notes

- CPG_Null_Runner/cpg_null_runner.py null N7 loads the synthetic patient generator, which was retired with chain v2; N7 reports DEFERRED until a v3 generator exists. The other nulls run without it.
- cmb_tools.py checks fields of the retired v2 bundle (patient_sky, foreign_detection, second_opinion); on a v3 bundle every entry reads NOT_RUN until the stage it checks is wired in.
- Kept out, and why (SOP section 2b): the class-era stage 10 panel, 3b panel, 3c panel, 11b archives and the toolkit NILC module N1 (each named in its row above), and null N7 of the null runner (its generator was retired).
- The class-era wrappers that called these modules inside the retired conductor (the stage 4.6 sky wrapper, the class-level NILC second opinion) were not carried over: they read retired files (class identity loci, beta scale maps). Each stage gets a v3 wrapper when it is commissioned.

Sources

manual/OM_v3_neutrophil_chain.md sha256 dc44057b8a79

doors/CHAIN_SEQUENCE.md sha256 8977095486f6

chain/TOOLKIT.md sha256 d4890053d11f